CO2 - AT1 - Analytical Problem Solving

Thresholding and intensity-Level slicing - PCB inspection.

Given

$r = [25, 60, 95, 130, 165, 200, 235]$

Solder joint intensity band : 90-170

a. Binary Thresholding

The transformation is:

$$s = \begin{cases} 255, & r \geq 128 \\ 0, & r < 128 \end{cases}$$

Applying threshold to each pixel:

Input	Output
25	0
60	0
95	0
130	255
165	255
200	255
235	255

Output : $s = [0, 0, 0, 255, 255, 255, 255]$

b. Intensity level slicing.

The transformation is:

$$s = \begin{cases} 255, & 90 \leq r \leq 170 \\ r, & \text{otherwise} \end{cases}$$

Output :

$s = [25, 60, 255, 255, 255, 200, 235]$

Input	Output
25	25
60	60
95	255
130	255
165	255
200	200
235	235

c. Number and percentage of Highlighted pixels.

Pixels within the band 90-170 are:

95, 130, 165

Number of highlighted pixels:

3

percentage: $\frac{3}{7} \times 100 = 42.86\%$

$\therefore$ **percentage = 42.86%**

d. Mean Intensity:

Original mean:

$$\bar{r} = \frac{25+60+95+130+165+200+235}{7}$$

$$\bar{r} = \frac{910}{7}$$

$$\boxed{\bar{r} = 130}$$

Sliced output Mean:

$$\bar{s} = \frac{25+60+255+255+255+200+235}{7}$$

$$\bar{s} = \frac{1285}{7} = 183.57$$

e. Intensity level slicing is more appropriate for isolating solder joints because it highlights only pixels in the specified 90-140 intensity band.

Binary thresholding highlights all pixels with intensity at least 128, including the pixel values 200 and 235, which lie outside the target band.

2. Piece wise -Linear Contrast stretching -X-Ray image enhancement.

Given

$(r_1, s_1) = (40, 20)$

$(r_2, s_2) = (140, 230)$.

input pixels:

$$r = [80, 70, 90, 110, 140, 200]$$

The transformation connects the points:

(0,0), (70,20), (140,230), (255,255)

a. Equations and slopes of the three segments.

the slope between two points is:

$$m = \frac{s_2 - s_1}{r_2 - r_1}$$

Segment 1: $0 \le r \le 70$

$$m_1 = \frac{20-0}{70-0} = \frac{20}{70}$$

$$m_1 = 0.2857$$

Equation:- $\boxed{s = \frac{2}{7} r}$

Segment 2: $70 < r \le 140$

$$m_2 = \frac{230-20}{140-70}$$

$$m_2 = \frac{210}{70}$$

$$\boxed{m_2 = 3}$$

using point slope equation

$$s - 20 = 3(r - 70) \Rightarrow \boxed{s = 3r - 190}$$

Segment 3: $140 < r \le 255$

$$m_3 = \frac{255-230}{255-140} \Rightarrow \frac{25}{115}$$

$$\therefore \boxed{m_3 = 0.2174}$$

Equation:

$$s - 230 = \frac{5}{23}(r - 140)$$

$$\therefore \boxed{s = 230 + \frac{5}{23}(r - 140)}$$

b. Calculate the transformed values

For $r = 30 \Rightarrow s = \frac{2}{7}(30) = 8.57$

$r = 70 \Rightarrow s = \frac{2}{7}(70) = 20$

$r = 90 \Rightarrow s = 3(90) - 190 = 80$

$r = 110 \Rightarrow s = 3(110) - 190 = 140$

$r = 140 \Rightarrow s = 230$

$r = 200 \Rightarrow s = 230 + \frac{5}{23}(200 - 140)$

$s = 243.04$

Final table

Input r	Transformed "s"
30	8.57
70	20
90	80
110	140
140	230
200	243.04

Transformed output:

[8.57, 20, 80, 140, 230, 243.04]

c. Band width and contrast Gain

Original diagnostic band:

$W_{original} = 140 - 70$

$= 70$

Transformed band:

$W_{trans} = 230 - 20$

$= 210$

contrast Gain:

$G = \frac{W_{trans}}{W_{original}} = \frac{210}{70} = 3$

∴ band expanded by a factor = 3

d. Identify compression and expansion

segment	slope	effect
0-70	0.2857	compression
70-140	3	Expansion
140-255	0.2175	compression

e. Does transformation improve visibility?

Yes, the diagnostic region from gray levels 70 to 140 is expanded from an output range of 20-230, making difference in intensity more noticeable.

Spatial Filtering - Impulse Noise Removal.

Given

3x3 neighborhood: $\begin{bmatrix} 52 & 55 & 58 \\ 54 & 255 & 57 \\ 53 & 56 & 59 \end{bmatrix}$ ∴ The center pixel, 255 is salt noise.

a. 3x3 Mean Filter

$$\text{Mean} = \frac{\text{Sum of all 9 pixels}}{9}$$

$$\frac{\text{Sum}}{\text{total}} = \frac{52+55+58+54+255+57+53+56+59}{9}$$

$$\text{Mean} = \frac{699}{9}$$

Mean = 77.67

b. Weighted - Average Filter

Given mask = $\frac{1}{16}\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$

Multiply each pixel by its corresponding weight:

weighted sum = 52 + 110 + 58 + 108 + 4(255) + 114 + 53 + 112 + 59

sum = 1686

Output = $\frac{1686}{16}$

Output = 105.37

c) Median Filter

Arrange all 9 pixel values in ascending order.

52, 53, 54, 55, 56, 57, 58, 59, 255

Total vales = 9 Median = 5th value

Median = 56

d. Absolute Error of Each filter

The noise-free reference is average of eight neighboring pixels.

$$R = \frac{52+55+58+54+57+53+56+59}{8}$$

$$R = \frac{444}{8}$$

$$R = 55.5$$

Absolute error:

$$E = |\text{Filter Output} - R|$$

Mean filter:

$$E_{mean} = |77.67 - 55.5| = 22.17$$

weighted-avg filter:

$$E_{weighted} = |105.375 - 55.5| = 49.875$$

Median filter:

$$E_{median} = |56 - 55.5| = 0.5$$

Comparision

Filter	Output	Absolute Error
Mean	77.67	22.17
weighted Avg	105.375	49.875
Median	56	0.5

e. Most Appropriate filter:

The median filter is most appropriate for removing salt and pepper noise while generally preserving edges. With less Error compared to the other methods.

Edge Detection - Sobel & Prewitt Gradient Analysis.

Given

3×3 region $A = \begin{bmatrix} 10 & 10 & 10 \\ 10 & 10 & 90 \\ 10 & 90 & 90 \end{bmatrix}$

a) Sobel Gradient G_x and G_y

Sobel G_x

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

Multiply G_x and A

$$G_x = (-10 \times 10) + (-20 \times 0 + 180) + (-10 + 90)$$

$$\boxed{G_x = 240}$$

Sobel G_y

$$G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

$$G_y = (-10 - 20 - 10) + (10 + 180 + 90)$$

$$G_y = -40 + 280$$

$$\boxed{G_y = 240}$$

b) Gradient Magnitude and Approximation

Exact gradient Magnitude

$$M = \sqrt{G_x^2 + G_y^2}$$

$$= \sqrt{240^2 + 240^2}$$

$$\boxed{M \approx 339.41}$$

Approximation

$M_{approx} = |G_x| + |G_y|$

$= 240 + 240$

$= 480$

Percentage error:

$$\% \text{ error} = \frac{|480 - 339.41|}{339.41} \times 100$$

% Error = 41.42%

c) Gradient Direction & edge Orientation

$$\alpha = \tan^{-1}\left(\frac{G_y}{G_x}\right) = \tan^{-1}\left(\frac{240}{240}\right)$$

$\alpha = 45°$

edge orientation $\perp$ to gradient

∴ **Orientation = 135°**

d) Prewitt Operator

$G_x = (-10 + 10) + (-10 + 90) + (-10 + 90)$

$G_x = 160$

$G_y = (-10 - 10 - 10) + (10 + 90 + 90)$

$G_y = 160$

Gradient Magnitude : $M = \sqrt{160^2 + 160^2} \approx 226.27$

$M_{approx} = 160 + 160 = 320$

Percentage error : 41.42%

e) Edge Detection using $M \geq 250$

Sobel detects an edge because its magnitude 339.41 exceeds 250. So it does not detect the edge because its magnitude 226.27 is below 250.

Global Thresholding - Iterative and Otsu Segmentation

a) Global Mean Intensity

$$m_G = \frac{\sum r_k n_k}{\sum n_k} = \frac{122}{32}$$

$$\therefore \boxed{M_G = 3.8125}$$

b) Iterative Global Thresholding

Initial threshold

$$T_0 = 3.8125$$

Class means

$$m_1 = \frac{17}{14} = 1.2143$$

$$m_2 = \frac{105}{18} = 5.833$$

Updated threshold:

$$T_1 = \frac{m_1 + m_2}{2}$$

$$T_1 = \frac{1.2143 + 5.83}{2} \Rightarrow 3.5238$$

since $|3.5238 - 3.8125| = 0.2887 < 0.5$, algorithm converges.

c) Otsu's method

Formula

$$\sigma_B^2 = P_1 P_2 (m_1 - m_2)^2$$

K	Between class variance
2	5.128
3	5.843
4	5.138

d. Optimum threshold & Binary Mapping

The max variance occurs at $\boxed{k=3}$

$$\boxed{k^* = 3}$$

Intensity	Binary output
0, 1, 2, 3	1 (Object)
4, 5, 6, 7	0 (Background)

Object pixels:

$$3+6+4+1 = 14$$

percentage : 0.14%

e) comparision and conclusion

- Iterative threshold = 3.5238
- Otsu threshold = 3

Both give similar threshold.

∴ histogram is reasonably suitable for global thresholding.